SNPoptimizer - User Manual

Updated on 2026-02-08

1. Overview

SNPoptimizer is a Shiny app to: - select a minimal SNP set that maximizes sample discrimination; - evaluate an already defined SNP set; - generate flanking sequences and Primer3 input files.

The app supports **HapMap** and **VCF** inputs (biallelic variants with `GT`).

2. Start the app

From the project folder:

```
shiny::runApp()
```

Alternative:

```
source("run.R")
```

Upload size limit is set to **1 GB**.

3. Discovery section

3.1 Input

- Upload HapMap (`.txt/.tsv/.hmp`) or VCF (`.vcf/.vcf.gz`).
- Set SNP heterozygosity threshold.
- Set `k` (number of SNPs to select).

3.2 Selection method

Two methods are available: - **Hybrid (greedy + local search)**: usually faster.
- **GA**: classical genetic algorithm with `popSize` and `maxiter`.

3.3 Reproducibility

You can enable a fixed seed (`useSeed`) for reproducible results.

3.4 Additional run (optional)

If duplicates remain, you can run a second search on duplicate samples only to add extra SNPs.

4. Discovery outputs

The app shows: - KPIs (filtered SNPs, selected SNPs, samples, fitness, duplicates); - method used (Hybrid/GA); - SNP discrimination curve; - selected SNP table; - remaining duplicate sample groups.

Available downloads: - selected SNP metadata; - selected genotype matrix; - discovery summary; - curve plot; - duplicated sample list.

5. Evaluation (given SNPs) section

This section evaluates a SNP list (CHR, POS) on: - a dedicated evaluation dataset, or - the dataset already loaded in Discovery.

Supported SNP-list formats: - file with `CHR` and `POS` columns; - free text, e.g. `chr1:12345` or `1 12345`.

Output: - requested/found SNP counts; - list of missing SNPs; - downloadable subset matrix.

6. Post-discovery: Flanking and Primer3

6.1 Supported FASTA

- `.fa`, `.fasta`, `.fa.gz`

6.2 Performance

If `samtools` is available and FASTA is not `.gz`, the app uses `faidx` for fast extraction.

6.3 Chromosome matching

CHR matching is robust for common naming variants (e.g. `1`, `chr1`, `SL4.0ch01`). If matching fails, the app shows a warning with FASTA header preview.

6.4 New skipped-SNP diagnostics

During flanking generation, some SNPs may be excluded (e.g. unresolved CHR, out-of-range POS). The app now: - shows a notification with skipped SNP count; - provides `Download skipped SNPs` (TSV) with skip reasons.

This prevents silent SNP loss in outputs.

7. Common issues and fixes

7.1 "Maximum upload size exceeded"

- Use the updated app version (1 GB limit);
- verify you are launching the app from the correct folder.

7.2 "No requested chromosomes were found in the FASTA"

- check assembly and chromosome naming consistency;
- verify required chromosomes exist in the FASTA;
- if available, use indexed FASTA (`samtools faidx`).

7.3 Flanking output has fewer SNPs than expected

- download `Download skipped SNPs` to see which SNPs were excluded and why.

8. Requirements

Main R packages: - shiny - GA - data.table - ggplot2

Optional external tool: - samtools (recommended for large FASTA files)

9. Method note (paper)

- **GA** mode remains available for GA-based workflows and reporting.
- **Hybrid** mode is a practical faster alternative for many datasets.
- In the paper, report which optimizer was used for each analysis.

10. Main project files

- `app.R`
- `ui.R`
- `server.R`
- `run.R`
- `install.R`